

1. Typst + Lilaq + Physica: Scientific Document Example

This single file demonstrates high-quality typesetting for mathematics, quantum mechanics, statistics, and finance-related visualizations.

1.1. Quantum Mechanics Equations

The time-independent Schrödinger equation in one dimension reads:

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x) \psi(x) = E \psi(x)$$

A common normalized Gaussian wave packet is:

$$\psi(x) = \frac{1}{(\sigma\sqrt{2\pi})^{\frac{1}{2}}} \exp\left(-\frac{x^2}{4\sigma^2}\right)$$

Using Dirac notation (imported from physica):

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \text{with} \quad |\alpha|^2 + |\beta|^2 = 1$$

The inner product is:

$$\langle\varphi||\psi\rangle = \int \varphi^*(x) \psi(x), dx$$

1.2. 2D Charts with Lilaq

1.2.1. Line Plots of Mathematical Functions

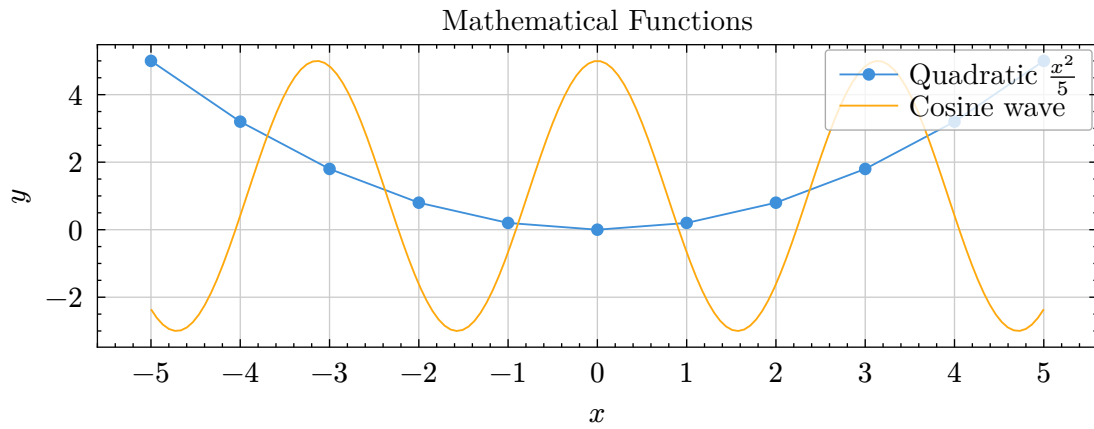


Figure 1: Line plots created with Lilaq. Functions are evaluated directly inside Typst.

1.2.2. Contour Plot (2D Wave-like Function)

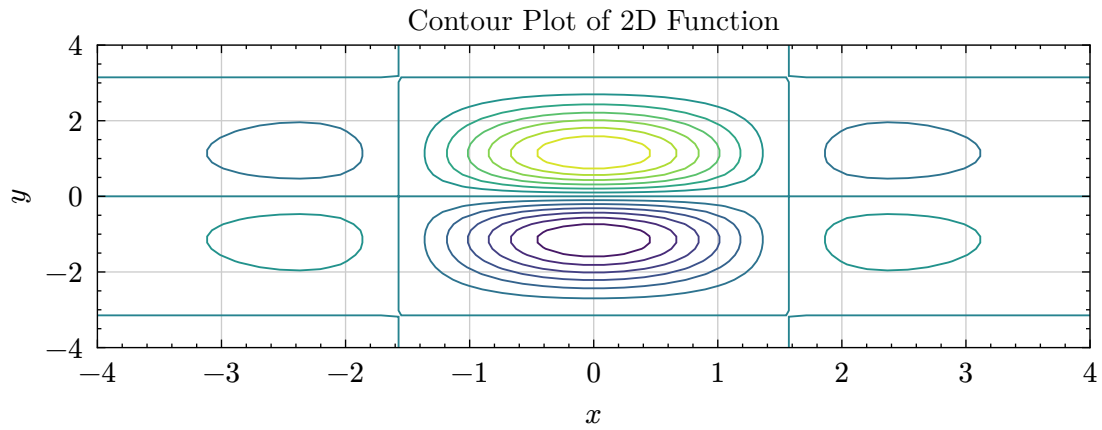


Figure 2: Contour plot using Lilaq. Useful for quantum mechanics probability densities or potential surfaces.

1.2.3. Quiver Plot (Vector Field)

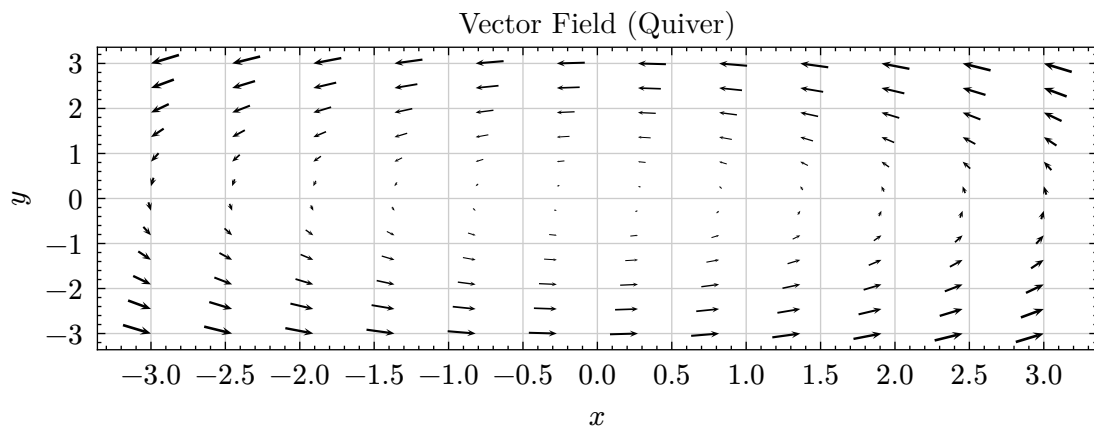


Figure 3: Quiver plot showing a rotational vector field with Lilaq.

1.3. 3D Surface Plot with plotsy-3d (Smaller Size)

A compact 3D surface of the function $z = \sin(x) \cos(y) \exp\left(-\frac{x^2+y^2}{8}\right)$.

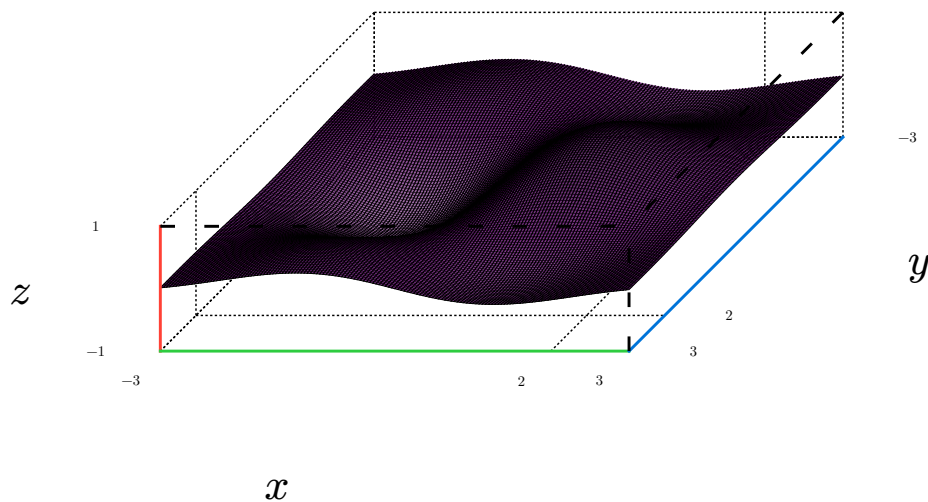


Figure 4: 3D surface plot with plotsy-3d